SKYWAVE™
AN ORBCOMM COMPANY



SC 1000
Hardware Guide (Partners)

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PREFACE

Purpose

This document is as an overview of the hardware specifications for the SC 1000.

Notation

Hardware components, labels, and figures in this document might not be exactly as shown and are subject to change without notice.

CAUTION: This safety symbol warns of possible hazards to personnel, equipment, or both. It includes hazards that will or can cause personal injury, property damage, or death if the hazard is not avoided.

Note: A note indicates information with no potential hazard. A note indicates points of interest or provides supplementary information about a feature or task.

Numbered lists indicate a series of steps required to complete a task or function.

Bulleted lists highlight information where order or sequence is not crucial.

Battery Safety Warnings

CAUTION: Always follow local disposal guidelines to properly dispose of the Lithium-ion battery and the device.

CAUTION: DO NOT throw the internal battery or the device into fire.

Environmental Protection

Waste electrical products should not be disposed of with household waste. Please recycle where facilities exist. Check with our Local Authority or retailer for recycling advice.

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1 OVERVIEW

The SC 1000 is a solar-powered, satellite, self-contained tracking solution principally designed for tracking and monitoring globally roaming assets. The SC 1000 interfaces with wireless sensors through Bluetooth Low Energy and integrated internal sensors to monitor the status and condition of the asset.

The device is powered through an internal battery that is recharged using an integrated solar panel.

Figure 1: SC 1000



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2 SPECIFICATIONS

2.1 Temperature

Parameter	Value
Operating Temperature Range	-30° to +70°C (-86° to +158°F)
Storage Temperature Range	-40° to +80°C (-40° to +176°F)
Battery Charging Temperature Range (nominal)	-20° to +50°C (-4° to +122°F)
Battery Discharge Temperature Range	-40° to +85°C (-40° to +185°F)

2.2 Temperature Sensor

The device measures internal temperature over the range of -40 to +85°C (-40 to +185°F) with a typical accuracy of 1.5°C.

The temperature sensor shuts down between readings to conserve power.

2.3 Electrical

2.3.1 Power

The device is powered through the internal battery that is recharged using an integrated solar panel.

2.3.2 Solar Panel (Battery Charging)

The solar panel wakes the device from ship mode. Exposing the solar panel to sunlight wakes the device.

The solar panel provides power to the battery charging system. The battery charging electrical specifications are:

Parameter	Min.	Max.	Units
Solar Open Circuit Voltage	4.84	-	V
Solar Power	400	-	mW
Charge Current	1.0	100	mA

2.3.3 Battery Electrical Specifications

The device includes a rechargeable long life, industrial grade 3.6V nominal battery able to supply the current necessary for all operational modes. The battery pack is encapsulated within the enclosure

An industrial lithium-ion battery provides power to the SC 1000.

Parameter	Min.	Typ.	Max.	Units
Capacity	330	-	-	mAh
Voltage (nominal)	-	3.6	-	V
Maximum Current Output	2.0	-	-	A
Maximum Charge Current	100	-	-	mA

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2.3.3.1 Battery Controller

Charge and discharge are controlled so that battery discharge is not allowed over the battery's safe discharge operating conditions.

Parameter	Typ.	Max.	Units
Charge Current	-	200	mA
Float Voltage	4.05	-	V
Charge Time	-	8	Hours
Over Discharge Voltage	2.5	-	V
Over Charge Voltage	-	4.2	V
Over Discharge Current	-	3.13	A
Short Circuit Current	-	6	A

2.3.4 Current Consumption

Current values are at nominal battery voltage and room temperature.

Parameter	Current (average)
Sleep	74 µA
Satellite Transmit (pulsed)	1346 mA
Satellite Receive	197 mA
Tracker (for example, searching for a GNSS fix)	36 mA

2.4 RF Specifications

2.4.1 OGx

Parameter	Value
Rx Operating Frequency	1525-1559 MHz
Modulation	OQPSK
Symbol Rate	2000, 8000, 16000 symbols / second
Polarization	RHCP
Tx Operating Frequency	1626.5-1660.5 MHz
Modulation	OQPSK
Symbol Rate	800, 1600, and 3200 symbols / second
Polarization	RHCP

2.4.2 Elevation Angle

The SC 1000 satellite antenna supports operation down to a minimum elevation angle of 20° without a ground plane.

Note: Use of a ground plane or elevated installation may improve antenna performance at elevation angles lower than 20°; however, performance in such scenarios is not guaranteed and may vary depending on environmental conditions.

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Note: For maritime installations, pitch / roll and yaw may affect the antenna's look angle especially on smaller crafts. When assets are deployed at higher latitudes with an elevation angle close to 20°, vessel motion can push the satellite in and out of view, affecting availability

2.4.3 BLE RF Interface

The device includes an internal BLE antenna.

2.5 GNSS

The GNSS module can be turned off to save energy. The major constellations include GPS, GLONASS, BEIDOU, and GALALEO.

The GNSS system includes an internal antenna. The 3D, aided position accuracy is <10 m CEP.

Table 1: GNSS Specifications

Parameter	Typical per Constellation
Time to First Fix[1]	
Cold Start	35 s
Warm Start	28 s
Hot Start	2 s
Sensitivity	
Tracking	-157 dBm
Hot Start	-156 dBm
Cold Start	-146 dBm
Accuracy	
Horizontal Position (CEP)	2.5 m

2.6 Memory

Parameter	Value
Non-Volatile Flash	16 MB, 100,000 write-erase cycles over its operating life.

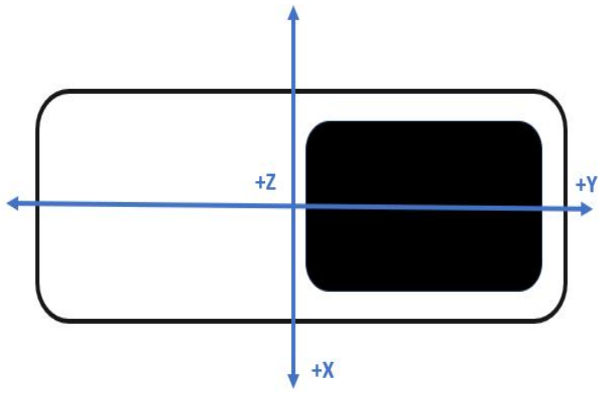
2.7 Accelerometer

The device has an accelerometer to detect motion, vibration, and shock. The device supports 16-bit signed acceleration data.

The following figure shows the orientation of the accelerometer.

[1] All satellites at -130 dBm

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The resolution and sensitivity of acceleration values reported by the accelerometer are dependent on the active gRange value.

Table 2: Supported Gravity Ranges and Device Resolutions

gRange	Acceleration Measurement Range	Resolution
2	±2 g	61.04 µg/LSB
4	±4 g	122.07 µg/LSB
8	±8 g	244.14 µg/LSB
16	±16 g	488.28 µg/LSB

Table 3: Supported Gravity Ranges and Device Sensitivities

gRange	Acceleration Measurement Range	Sensitivity
2	±2 g	16384 LSB/g
4	±4 g	8192 LSB/g
8	±8 g	4096 LSB/g
16	±16 g	2048 LSB/g

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2.8 Mechanical

Parameter	Value
Weight	519 g (18 oz)

The device's mechanical dimensions are shown in mm [inches].

Figure 2: Top View

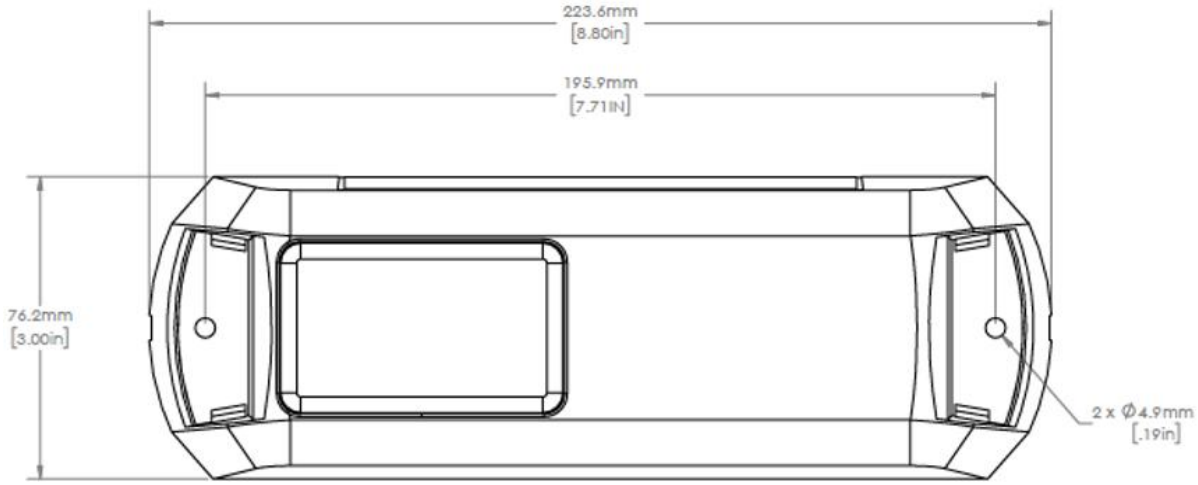
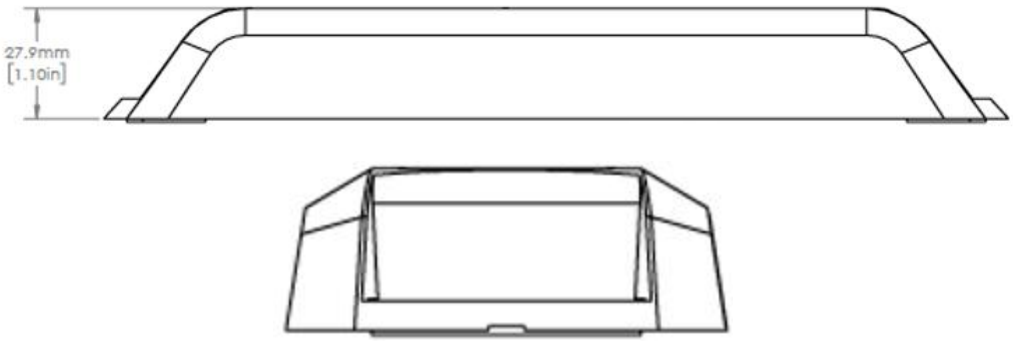


Figure 3: Side View



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2.9 Environmental

Parameter	Description
Humidity	The device meets all its specifications during exposure to 90% relative humidity at +70°C (158°F), per the test methodology of SAE J1455, section 4.2.3 (8-hour humidity cycle per figure 4a).
Vibration	The device meets all its specifications during exposure to random vehicular vibration levels per AAR-S-9401 3.2.4.2.
Mechanical Shock	The device meets all its specifications after exposure to positive and negative saw tooth shock pulses with peaks of 40 G and durations of 11 ms as specified in MIL-STD-810H, section 516.8, Procedure I, section 2.3.2c.
Thermal Shock	The device meets all of its specifications after a thermal shock test as detailed in SAE J1455, section 4.1.3.2.
Thermal Cycle	The device meets all of its specifications after a thermal cycle test as detailed in SAE J1455, section 4.1.3.1, Figure 2A.
Drop Test	The device meets all its specifications after a handling drop test as specified in SAE J1455, section 4.11.3.1.
Altitude	The device meets all specifications after a nonoperating 12.2 km (7.5 miles) altitude test as detailed in SAE J1455, section 4.9.3, except with an ambient temperature of -40°C (-40°F).
Salt Spray Atmosphere	The device meets all of its specifications after a salt spray test as detailed in SAE J1455, section 4.3.3.1.
Immersion	The device meets all of its specifications after an immersion test as detailed in SAE J1455, section 4.3.3.2.
Steam Cleaning and Pressure Washing	The device meets all of its specifications after a steam cleaning and pressure wash test as detailed in IPx9K per ISO 20653 Table 4, Table 7, and Figure 7.
Fungus	The device meets all of its specifications after a fungus test as detailed in SAE J1455, section 4.6.3.
Exposure to Chemicals and Oils	The device meets all of its specifications after a light to moderate splash test as detailed in SAE J1455 section 4.4.3.2, for the following chemicals: Window Washer Solvent Antifreeze Water Mixture Brake Fluid Dust Control Agents (magnesium chloride) Moisture Control Agents (calcium chloride) Engine Degreaser Gasoline Fuel Additives Battery Acid Kerosene Spray Paint Muriatic Acid Soap and Detergents Axle Grease Diesel Exhaust Fluid Diesel Fuel Engine Oils and Additives Ether Rear Axle Oil Alcohol Waxes Freon Paint Stripper Ammonia
IP67	The device meets all of its specifications after dust and water ingress testing as detailed in IEC 60529 section 13.4, 13.6, 14.2.7, and 14.3.

Parameter	Description
ESD	(Enclosure) All electrical interfaces operate normally after being subjected to 6 kV ESD contact discharge per IEC 61000-4-2 human body model, level 3.
External source of heating or cooling	**DO NOT** mount this device on or next to any external sources of heating or cooling.

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3 COMPLIANCE

CE Mark (Europe)

- RED 2014/53/EU

  Declaration of Conformity

  Hereby, ORBCOMM declares that the radio equipment type SC 1000 is in compliance with Directive 2014/53/EU.

  The EU DOC (Declaration of Conformity) is either included in the packaging or can be found at the following link: http://www2.orbcomm.com/eudoc.

- EN 60079-0: 2018, Explosive atmospheres – Part 0: Equipment – General requirements.
- EN 60079-11, 2012, Explosive Atmospheres Part 11: Equipment protected by intrinsic safety.
- IEC 60529 rev 2.1, Degrees of protection provided by enclosures (IP code), February 2001.

ISED (Canada)

- Contains IC ID: 11881A-BTCT1 & 10224A-201906EG21G.
- Compliance Statement

  This device complies with Industry Canada license-exempt RSS standard(s). Operation is subject to the following two conditions: (1) This device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

  Le present appareil est conforme aux CNR d'Industrie Canada applicable aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes: (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, meme si le brouillage est susceptible d'en compromettre le fonctionnement.

- ICES-003, Issue 4, Interference-Causing Equipment Standard – Digital Apparatus, February 2004.
- RSS-102, Issue 4, Radio Standards Specification - Radio Frequency (RF) Exposure Compliance of Radiocommunication Apparatus (All Frequency Bands), March 2010.

FCC (United States of America)

- Contains FCC ID: XGS-BTCT1 & XMR201906EG21G.
- FCC Compliance Statement

  This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:

  (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

- FCC Part 15, Radio Frequency Devices.
- FCC OET Bulletin 65 Edition 97-01, Evaluating Compliance with FCC Guidelines for Human Exposure to Radio frequency Electromagnetic Fields, August 1997.

Ingress Protection

Device enclosure: IP67 and IP69k

HERO

Hazard of Electromagnetic Radiation to Ordnance

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ATEX

Refer to APPENDIX A ATEX Safe Use Guidance for details.

RoHS

Restriction of Hazardous Substances (RoHS)[1]

IEC 60945

Maritime navigation and radio communication equipment and systems – general requirements

Country Approval

Australia, Austria, Argentina	Italy	Peru, Poland
Belgium, Brazil	Japan	Saudi Arabia, South Africa, Spain
Czech Republic, Canada, Chile, Colombia	Mexico	UAE, United Kingdom , United States of America
Finland, France	Netherlands, New Zealand	
Germany	Oman	

[1]European Union's (EU) Directive 2002/95/EEC "Restriction of Hazardous Substances" (RoHS) in Electronic and Electrical Equipment. For Chinese and European RoHS directives.

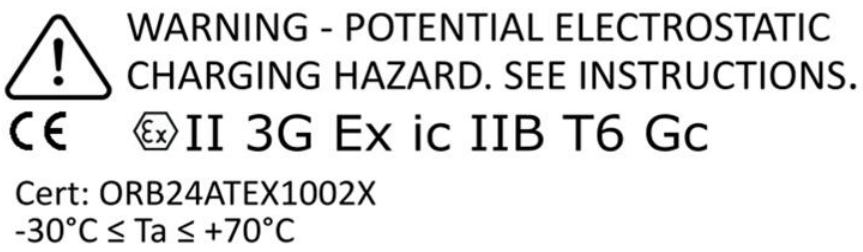
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APPENDIX A ATEX SAFE USE GUIDANCE

This section provides information, specific to the ATEX Category 3G certification, for safe use of the SC 1000 device.

Marking Label Information

The SC 1000 device is provided with marking label information specific to ATEX Category 3G certification:



Applicable Standards

A list of standards with which the equipment is declared to comply:

- Equipment or Protective System Intended for use in Potentially Explosive Atmospheres Directive (ATEX) 2014/34/EU
    - EN 60079-0:2018, Explosive atmospheres–Part 0: Equipment – General requirements
    - EN 60079-11:2012, Explosive atmospheres–Part 11: Equipment protection by intrinsic safety "i"
- Radio Equipment Directive (RED) 2014/53/EU, EN 62311: 2020, EN 62368-1: 2020+A1:2020
- Radio Equipment Directive (RED) (EMC) 2014/53/EU, EN 301 489-1 V 2.2.3(2019-11), EN 301 489-17 V 3.2.4 (2020-09), EN 301 489-52V 1.1.2(2020-12), EN 55032:2015/A11:2020, EN 55035: :2017/A11:2020
- Radio Equipment Directive (RED) (Radio Spectrum) 2014/53/EU, EN 301 511 V12.5.1, EN 301 908-1 V13.1.1, EN 301 908-2 V13.1.1, EN 301 908-13 V13.1.1, EN 303 413 V1.1.1, EN 300 328 V2.2.2 (1019-07)
- Restriction of Hazardous Substances Directive (RoHS) 2015/863/EU Article 4.1, EN50581:2012

Electrical Parameters

Parameter	Nominal	Units
Voltage	4.2	V
Current	45	mA
Power Dissipation	185	mW
Energy Harvesting	480	mW

Specific Conditions of Use:

There is a risk of potential electrostatic charge build-up on the plastic enclosure. The instructions shall be referenced for safe use and techniques to reduce the potential for electrostatic charge build-up:

In Zone 2 Hazardous Locations, insulating solid materials are allowed to be used if charging processes capable of generating hazardous potentials are unlikely to occur during normal operation including maintenance and cleaning. Synthetic fabrics used in cloths for cleaning or wiping can develop sufficient static electric charge to produce discharges capable of igniting solvent vapors. Typically, charge generation increases with the speed and vigor of the wiping action. The material being cleaned or wiped, if insulating, also can accumulate sufficient charge to produce an incendive discharge. Cotton or synthetic fabric treated with a static dissipative compound may be

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required if static electric charge generation needs to be controlled, especially if flammable insulating solvents are being used for cleaning or wiping.

"WARNING – Clean with a damp cloth only and allow to dry naturally"

Special Instructions

The SC 1000 device is meant for fixed installation.

The SC 1000 device does not require any maintenance.

The SC 1000 is not intended to be mounted on or used next to any external sources of heating or cooling.

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